COMP 1787 Requirements Management

Professional Issues

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Introduction

- The need for a professional approach in IT
- What is a Profession?
- What is a Professional Society?
- The Computing Professional Institutes
- Requirements of Membership
- Grades of Membership
- Benefits of membership
- Codes of Conduct and Practice

The Need for Professionalism in Information Systems Development

- 25% of all large scale developments cancelled

Scientific American 1994

- 2009 estimated cost of failure $6.2 Trillion worldwide, $1.2 Trillion in US, $200 Billion in UK.

ZDNET: Worldwide cost of Failure 2009

- €142 Billion cost EU on failed IT projects in 2004

BCS study on IT project failure June 2008

- 15% of all S/W projects fail to deliver anything at all
  62% of companies had runaways in last 5 years
  30% expecting one now

KPMG report for DTI

- Almost 33% of the projects failed altogether
  Almost 75% of IT projects exceeded their budgets & schedules

Gartner / Standish 1998

- About 40% of developments fail or are abandoned
  About 80% of systems are delivered late and over budget

OASIG 1998

- 31% of projects cancelled before completion
  88% of projects exceed deadline, budget or both
  For every 100 starts, there are 94 restarts
  Average cost overrun – 189%
  Average schedule overrun – 222%

Standish Report 2002

- Annual cost of poor quality software to the U.S. economy is $60bn and €60bn to the E.U.
  - US National Institute for Science & Technology, 2004
- KPMG report for the 2005 survey: 49% of participants experienced at least one project failure in the last year.
- KPMG report for the 2010 survey: 70% of participants experienced at least one project failure in the last year.
- KPMG report for the 2019 survey:19% Of organisations deliver successful projects, at least most of the time
- 35% of organisations are likely to deliver projects that are on budget
  - (Source: https://www.ipma.world/assets/PM-Survey-FullReport-2019-FINAL.pdf)

- COVID Test and Trace (2020) A total of £10bn was allocated
  Primary care had been ‘bypassed totally’ in the government’s strategy to tackle the virus
  System ‘a disaster and a national shame’
  (source:https://www.gponline.com/failure-involve-gps-covid-19-test-trace-a-disaster-national-shame/article/1698043)

Failure of software ranges from mild inconvenience to costing lives
(failure always costs money - customers if not suppliers).

IT as a Profession?

- Even though many projects are successful – is this level of performance acceptable ?
- Would a situation in other branches of engineering be accepted if, for example :
  - 4 bridges in 10 failed ??
  - 4 out of 10 miles of new highway were unusable ??

The Role Of The Professional

- Professionalism suggests:
  - Mastering an extensive body of knowledge
  - Providing an important service in order to advance or preserve a particular social value
  - Exercising monopolistic control over relevant services to society
  - Accepting a code of professional conduct that spells out professional responsibilities.

- Which of these applies to people working in the computing industry?

Mastering An Extensive Body Of Knowledge

- Most people working in the computing industry have experience from university and/or work experience and practice.
- However, it is not compulsory.
- there is a need for better focus on the social impact of computerised information systems.
- Failure to do this will lead to practitioners with little or no understanding of the ethical issues.
- As information technology pervades our lives, this becomes more and more critical.

Providing A Service To Advance Or Preserve A Particular Social Value

- Does not exist in the computing field as it does in other professions.
- Moor's "logically malleable" view of computing
  - IT can be massaged into any use we want

Exercising Monopolistic Control Over Relevant Services To Society

There is currently no licensing of computer practitioners

There is currently no mechanism for stopping a computer practitioner to continue to practice

Accepting A Code Of Professional Conduct

Professional bodies have a number of different codes, but computer practitioners are not strongly bound to them

Computer professionals can break guidelines at will with little or no comeback.

“Computing has matured to a point where it has standards, methods, and techniques which, if employed, would reduce the likelihood of many computing disasters. The world will no longer accept the view that bugs arise in programs by Spontaneous Generation. The bugs (errors) were put there by people who should know better. The failure to deliver quality products is unethical, even in computing”

Don Gotterbarn

Professional Approach

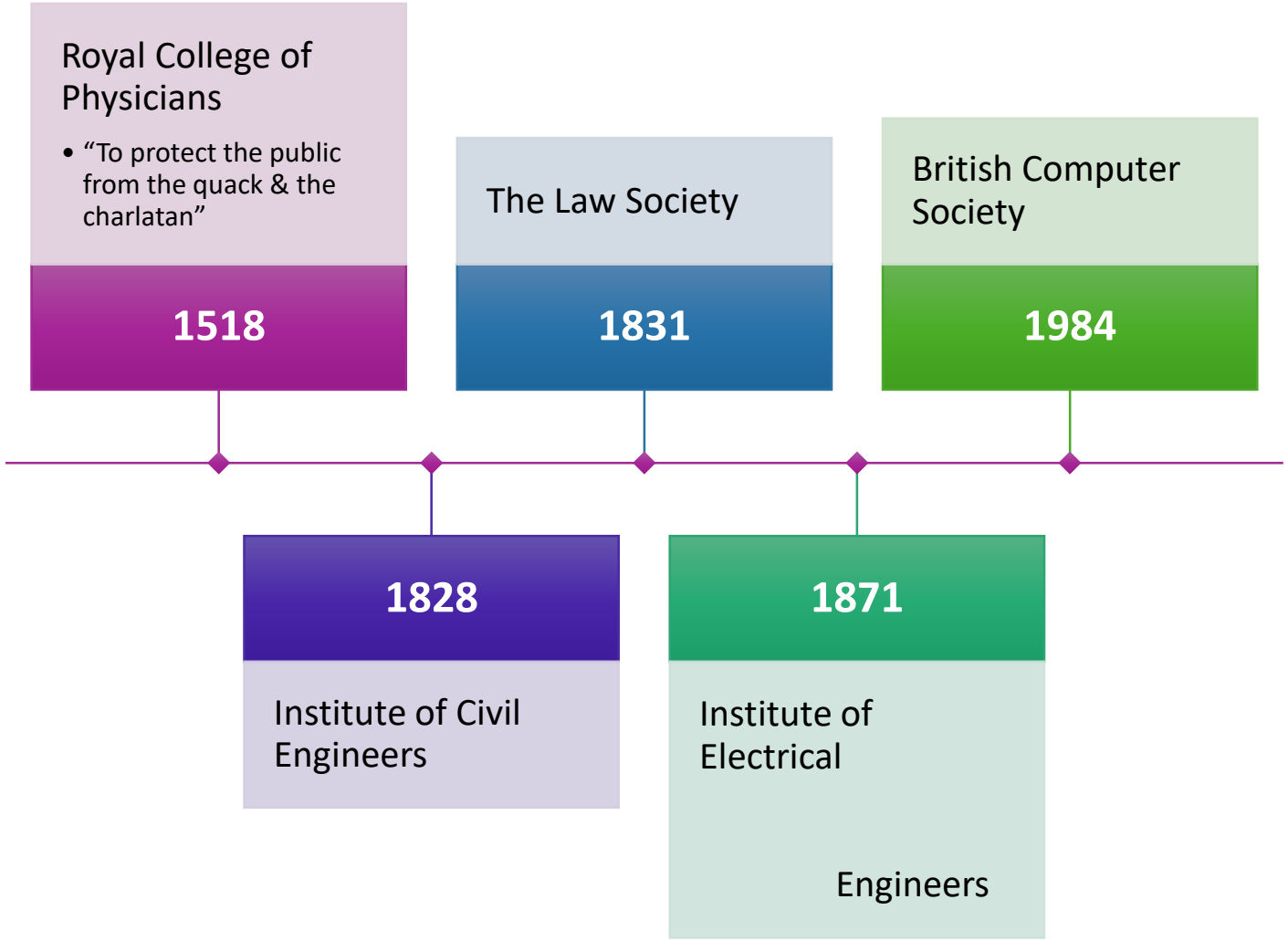
Professions such as medicine and Law are distinguished from other occupations by the period of training and experience required in order to qualify as a practitioner

They are also regulated by professional societies

- Separate legal identity
- Control entry into the profession
- Run by a college of peers
- Sets and enforces standards of professional behaviour and competence

Law is REACTIVE
Professions have a PROACTIVE role

Chartered Professions The Royal Colleges



A Profession

A Skilled Occupation
Within an Organised Framework:

- An Accepted Body of Knowledge
- Structured Training
- Skills Maintenance and Development
- A Code of Conduct and Practice

Professional Bodies Created:

- By Royal Charter
- By Statute

What does a Professional Body do?

"Professional associations are the bodies that represent our collective voice, that set standards for the profession, that encourage discussion of good practice, and that monitor training to ensure the responsible development of the profession."

John Leighfield, CBE, Chairman, RM plc and IMIS President

What does a Professional Body do?

- Represents the profession on issues of importance
- Leads formulation of national and international standards
- Sets standards for qualifications, education & training
- Maintains registers of Expert Witnesses, Consultants etc.
- Monitors professional conduct
- Provides dispute resolution services
- Advances and disseminates knowledge
- Recognises achievement
- Provides services to members
- Offers members opportunities for networking

- “In the software industry I have seen people who have taken a short programming course and become "software engineers". Now, you try taking your degree in computing or software and doing an "architectural appreciation course" and calling yourself an Architect…. or a six-month first aid course and call yourself a [Medical] Doctor. There are laws against this but you will have to be a barrister to defend yourself in court. This is because it has been deemed dangerous to have unqualified people as Architects, Lawyers, Doctors, Civil (structural) Engineers, Gas fitters etc. However, here is no virtually no restriction on embedded engineers no matter how safety critical the work.”

C. Hills, *A Standard Individual: A Licensed Engineer*, at http://www.phaedsys.demon.co.uk/chris/sweng/ESE/03-003esestda.pdf, 2004

Statutory Bodies and Licensed Professionals

Some professions are regulated by statute:

- Legally must hold a license to practice
- Must renew practicing certificate annually
- Subject to assessment of competence at regular intervals:
  - Demonstrate maintenance of knowledge and expertise through CPD/PDS
  - In some fields must resit exams every few years

Computing/Software Engineering is ***not*** currently a statutory licensed profession in the U.K.

Very few licensed engineering disciplines in U.K. includes:

- Railway Signal Engineers
- Aircraft Maintenance Engineers
- Pressure Vessel Design
- Reservoir Design and Inspection
- Quarry Management

Example

- In June 1998 Texas established SW Engineering as a recognised and licensed profession.
- You get the right to call yourself an Engineer and can offer software engineering services for gain. This is for any software for “engineered systems including embedded systems, real-time systems, mechanical devices, electrical devices and power systems”.
- The requirements are a suitable degree, 16 years experience and references from 9 people 5 of which must be Licensed Engineers.
- Computing is not currently a licensed profession in the U.K.

The Range of Computing – Which Professional Body?

- Business Process Engineering/Soft Systems — **IMIS**
- Information Systems Analysis and Design
- Multimedia and HCI Design
- Systems Development — **BCS**
- Programming, Testing
- Embedded systems
- Networking, Machine & Comms architectures
- Electronics, VLSI — **IET**

Institute for the Management of Information Systems

- Formally Institute of Data Processing Management (IDPM) restructured and changed name 1997
- IMIS is the leading professional association in the Information Systems management sector focuses specifically on the practical application and management of information systems within our society

BCS

- “The Chartered Institute for IT”
- Learned society and professional institute
- Formed in 1957
- Royal Charter in 1984
- Nominated Engineering Institution 1990

BCS

The Chartered Institute for IT
Enabling the information society

The BCS Royal Charter

"The objects for which the society is hereby constituted shall be to promote all aspects of computing and the use of computers and to advance knowledge and education therein, and to provide for the good of the public at large and the protection of individuals."

Elizabeth the Second by the Grace of God of the United Kingdom of Great Britain and Northern Ireland and of Our other Realms and Territories Queen, Head of the Commonwealth, Defender of the Faith.

TO ALL TO WHOM THESE PRESENTS SHALL COME, GREETING!

Whereas an humble Petition has been presented unto Us by the Company incorporated in the year of our Lord One thousand nine hundred and fifty seven under the Companies Act 1948 as a company limited by guarantee and not having a share capital and known as "The British Computer Society" (hereafter called "the Company") praying that We should constitute a Corporation incorporated by Our Royal Charter to acquire and take over the assets and carry on the activities of the Company in succession thereto and with such provisions as should seem to Us right and suitable:

And Whereas We have taken the said Petition into our Royal Consideration and are minded to accede thereto:

Now Therefor Know Ye that we by virtue of Our Prerogative Royal and of Our special grace, certain knowledge and mere motion have granted and declared and do hereby for Us, Our Heirs and Successors grant and declare as follows:

1. The members of the Company and all such persons as may hereafter become members of the Body Corporate hereby constituted shall forever hereafter be one Body Corporate and Politic by the name of "The British Computer Society" (hereinafter referred to as "the Society") and by the same name shall have perpetual succession and a Common Seal, with power to break, alter and make anew the said Seal from time to time at their will and pleasure and by the same name shall and may sue and be sued in all Courts and in all manner of actions and suits, and shall have power to do all other matters and things incidental or appertaining to a Body Corporate.

2. The objects for which the Society is hereby constituted shall be to promote all aspects of Computing and the use of computers and to advance knowledge and education therein, and to provide for the good of the public at large and the protection of individuals. In this our Charter and the Bye-laws the expressions "Computing" shall include matters concerned with the furtherance of computer science and technology and the design and development of computing systems and applications, and "computer" shall include any form of computing system. Provided that, in pursuing these objects, insofar as they may be similar to those of existing organisations, the Society shall use its best endeavours to co-operate with them and to ensure that its activities are complementary to those of such organisations.

3. In furtherance of its objects (but not further or otherwise and only to the extent to which the same may lawfully be exercised by a body having exclusively charitable objects) the Society shall, subject to this Our Charter and the Bye-laws, have the following powers:-

(a) to establish and maintain appropriate standards of education and experience for persons engaged in the profession of Computing or entering upon courses of study in Computing and allied subjects; and to undertake, supervise or encourage the education and training in all matters relevant to the advancement of Computing of persons who are engaged in or likely to be engaged in the practice of Computing paying due regard to the provisions of courses of instruction by other organisations;

(b) to establish and maintain a sound ethical foundation for the use of computers, data handling and information technology systems; and to adopt any lawful means conducive to the maintenance of a high standard of professional skill and conduct amongst members of the Society;

(c) to maintain a register of persons qualified in Computing by admission to one of the classes of corporate membership of the Society and to enrol persons not so qualified as non-corporate members;

The Institute Engineering and Technology

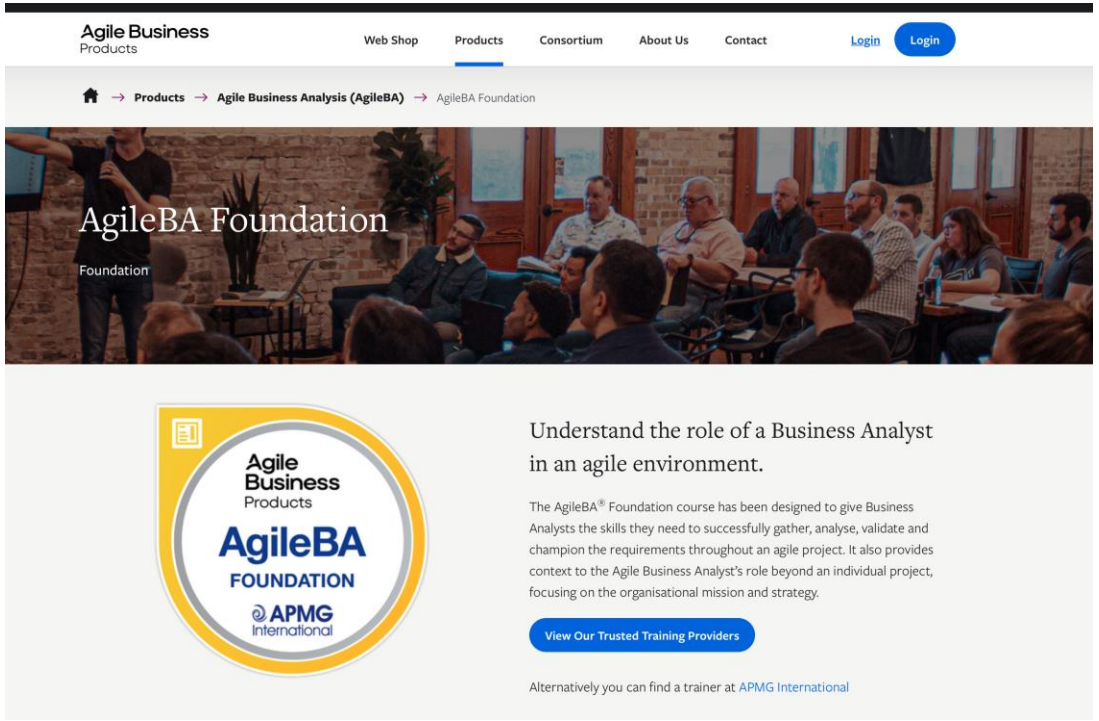
- Founded 1871, Royal Charter 1921
- Formally the IEE and IIE
- Represents the profession of electrical, electronic, manufacturing and systems engineering and related sciences.
- IET is the largest professional engineering society in Europe
- Lead Engineering Council Institute

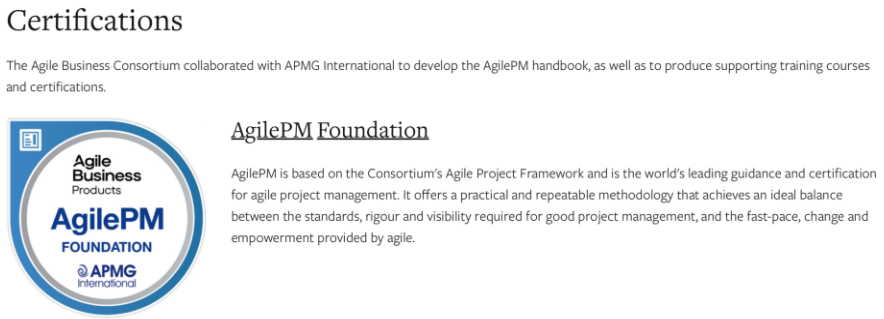
The Agile Business Consortium

- The Agile Business Consortium is an independent professional body dedicated to advancing business agility worldwide in accordance with a defined set of professional standards and a code of practice. It does this by:
- Supporting the development of professionals through pathways of formal CPD, informal learning and validated certification
- Providing access to: relevant, trustworthy, content; a range of events and special interest communities; a network of like-minded professionals
- Commissioning and engaging in research, thought leadership, and market-wide collaborations to extend knowledge and develop good practice

https://www.agilebusiness.org/about.html

The Agile Business Consortium



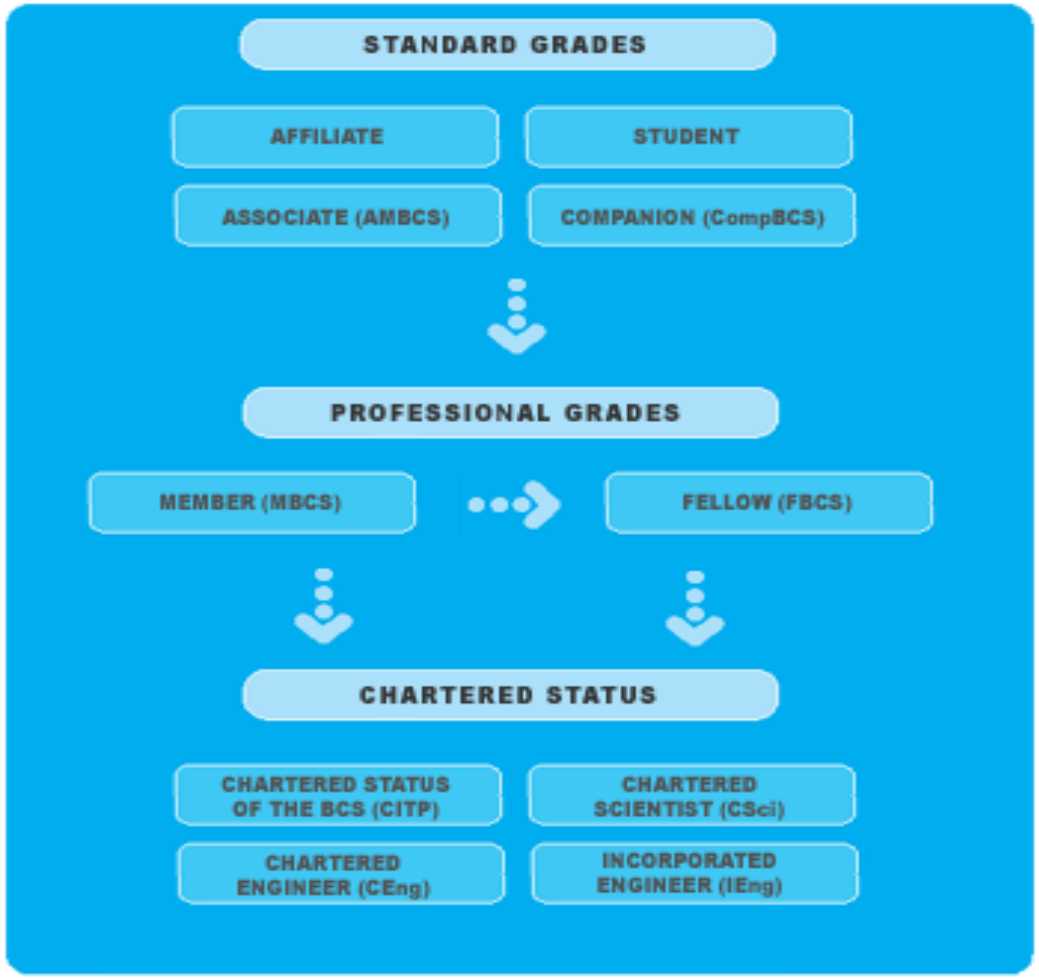


Membership of a Professional Society

- Requires:
  - Education
    - Society sets its own exams
    - Society accredits courses meeting standards
  - Experience
    - Typically, 5 years post-graduation
    - Continuous Professional Development schemes
  - Professionalism
    - Referees and Professional review
    - Adherence to code of conduct

Professional Membership Grades

- Three levels:
  - Standard Grades
  - Professional Grades
  - Chartered Status
- Based on:
  - Education
  - Experience
  - Professionalism

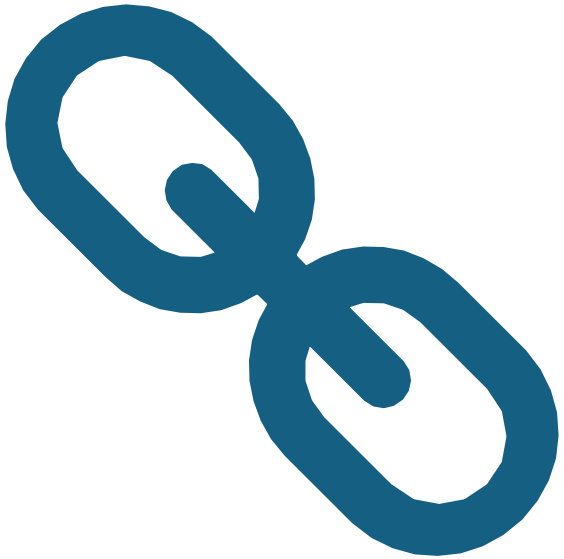


Chartered Professional Status

- **CITP**
  - Corporate membership + ~5 years appropriate experience + Professional review
- **I.Eng. & C.Eng.**
  - Chartered professional, meets Engineering Council SARTOR Industry Structure Model requirements
- **C.Sci**
  - Chartered Scientist - CITP + postgraduate science degree
- **EurIng**
  - FEANI registered Ingénieur - Chartered Engineer + 15 years appropriate experience at appropriate level

Grades of BCS Membership

- The Standard Grades:
- **Affiliate**
  - open to all – agree to adhere to codes of conduct
- **Student Member**
  - on an approved course of study
- **Associate Member (AMBCS)**
  - First step to full professional membership
  - non-accredited I.T. degree plus 1..5 years' experience



BCS Membership Grades Cont'd

- The Professional Grades:
- **Member (MBCS)**
  - Accredited degree
- **Fellow (FBCS)**
  - Senior member of the profession with established reputation of eminence or authority
- Other:
  - Companion (CompBCS)
  - Distinguished Fellow

Professional Networks

Branches

- A regional network covering the UK and a number of overseas locations
- Meetings, presentations, summer schools etc.
- Network with professionals working in your area
  – see http://www.londoncentral.bcs.org/events/forthcoming-events.php

Specialist groups

- A technical network covering a wide range of areas
- Meetings, newsletters, conferences, etc.
- Meet and network with professionals working in your technical area

The Young Professional Group

What's In It For Students

- Free Books & other publications
  - *Computer Bulletin and Journal*
  - *Computing*
  - *BCS review & directory*
  - *Inside Careers guide*
- Up to 25% discount on a range of books and publications from the BCS and others
- Other discounts
  - Conferences and other Events
  - Financial and Insurance Services
  - Hotel, Travel and Holiday Services

Professional Codes of Conduct and Practice

Members of a Professional Society agree to meet certain duties and responsibilities towards the public, the profession, employers and customers. *e.g.* Hippocratic Oath

- BCS code of conduct/code of practice
- IMIS code of ethics
- ACM/IEEE code of ethics
- Other professional codes…

Codes of Conduct

For example, **BCS Codes of Conduct** and **Codes of Practice.**

- Covers duties of care, conflicts of interest, competence, confidentiality and legal compliance.
- Encourages awareness and understanding of issues relating to the field of information systems.
- Required condition of membership
- Applies to members of all grades, including students
- Maintained by the BCS Ethics Panel.
- Enforced by the Society's disciplinary procedures.

Consider this Issue…….

- A software consultant is negotiating a contract with a local community to design their traffic control system.
- He recommends they select the TCS system out of several available systems on the market.
- The consultant fails to mention that he is a major stockholder of the company producing TCS software.

- **What is the most effective professional approach to address this situation?**

`Consider https://www.acm.org/code-of-ethics

ACM Guidelines for a Professional Approach

- According to the 2.5 ACM code of conduct guidelines- computer professionals must "*strive to be perceptive, thorough and objective when evaluating, recommending, and presenting systems descriptions and alternatives.”*
- *It also says that imperative 1.3 implies a computer professional must be honest about "any circumstances that might lead to conflicts of interest."*

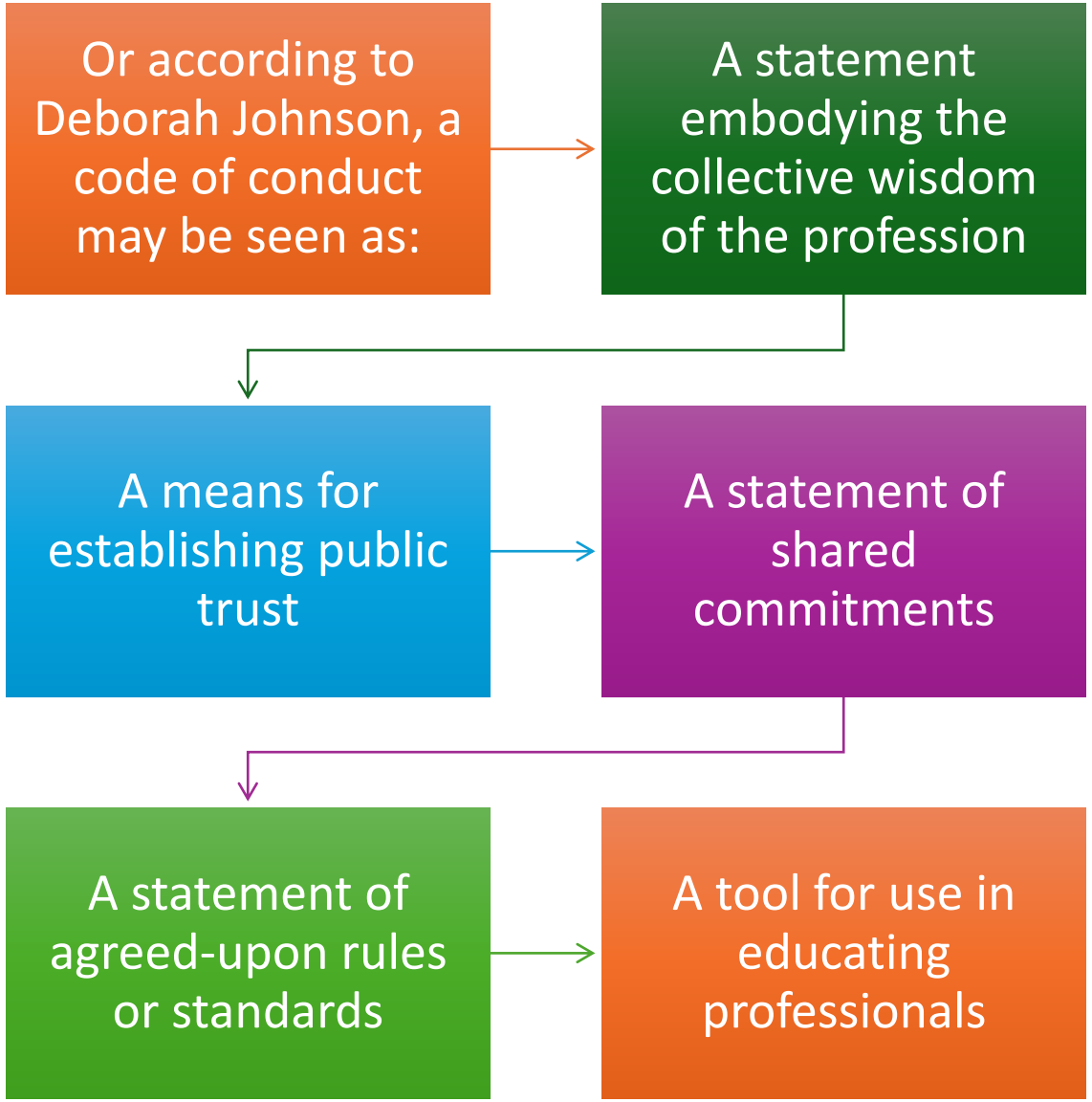
  Because of the special skills held by computing professional it is their responsibility to ensure that their clients are fully aware of their options and that professional recommendation are not modified for personal gain.

Source: https://dl.acm.org/doi/pdf/10.1145/151220.151231

What a Code Should Do

- Inspire
- Educate
- Guide
- Provide accountability
- Enforce

A Code of Conduct



What a Code is not

Not the law

Not a complete ethical framework or algorithm

Not an exhaustive checklist

BCS Code of Conduct

- Code sets out the professional standards required by the Society as a condition of membership.
- The Code governs your personal conduct as an individual member of the BCS and not the nature of business or ethics of the relevant authority. It will, therefore, be a matter of your exercising your personal judgement in meeting the Code's requirements.
- NOT academic philosophical study
- BUT Practical issues - e.g.
  Dealing with personal uncertainties
  Dealing with conflicts of opinion
  What choice should I make ?
  Is this action unfair ?

- Ref. Duncan Langford Practical Computer Ethics

BCS Code of Conduct - Four Sections

- **Public Interest** – account of the public interest and rights and interests of third parties, *e.g.* aware of relevant legislation
- **Duty to Relevant Authority**, *e.g.* Employers and Clients – Work carried out to proper standard and practitioners must not exploit position to gain personal advantage
- **Duty to the Profession** – Uphold the profession as a whole and support other members, *e.g*. help junior members develop
- **Professional Competence and Integrity** – Keep knowledge and expertise up to date

BCS Code of Conduct

The Public Interest

1. You shall carry out work or study with due care and diligence in accordance with the relevant authority's requirements, and the interests of system users. If your professional judgement is overruled, you shall indicate the likely risks and consequences.
2. In your professional role you shall have regard for the public health, safety and environment.
3. You shall have regard to the legitimate rights of third parties.
4. You shall ensure that within your professional field/s you have knowledge and understanding of relevant legislation, regulations and standards, and that you comply with such requirements.
5. You shall conduct your professional activities without discrimination against clients or colleagues
6. You shall reject any offer of bribery or inducement.

BCS Code of Conduct Duty To Relevant Authority

You shall avoid any situation that may give rise to a conflict of interest between you and your relevant authority. You shall make full and immediate disclosure to them if any conflict is likely to occur or be seen by a third party as likely to occur.

You shall not disclose or authorise to be disclosed, or use for personal gain or to benefit a third party, confidential information except with the permission of your relevant authority, or at the direction of a court of law.

You shall not misrepresent or withhold information on the performance of products, systems or services, or take advantage of the lack of relevant knowledge or inexperience of others.

BCS Code of Conduct Duty To The Profession

You shall uphold the reputation and good standing of the BCS in particular, and the profession in general, and shall seek to improve professional standards through participation in their development, use and enforcement.

You shall act with integrity in your relationships with all members of the BCS and with members of other professions with whom you work in a professional capacity.

You shall have due regard for the possible consequences of your statements on others. You shall not make any public statement in your professional capacity unless you are properly qualified and, where appropriate, authorised to do so. You shall not purport to represent the BCS unless authorised to do so.

You shall notify the Society if convicted of a criminal offence or upon becoming bankrupt or disqualified as Company Director

BCS Code of Conduct Professional Competence and Integrity

You shall seek to upgrade your professional knowledge and skill, and shall maintain awareness of technological developments, procedures and standards which are relevant to your field, and encourage your subordinates to do likewise.

You shall not claim any level of competence that you do not possess. You shall only offer to do work or provide a service that is within your professional competence.

You shall observe the relevant BCS Codes of Practice and all other standards which, in your judgement, are relevant, and you shall encourage your colleagues to do likewise.

You shall accept professional responsibility for your work and for the work of colleagues who are defined in a given context as working under your supervision.

ACM/IEEE Code of Ethics and Professional Practice

Public - Software engineers shall act consistently with the public interest.

Client And Employer - Software engineers shall act in a manner that is in the best interests of their client and employer, consistent with the public interest.

Product - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

Judgment - Software engineers shall maintain integrity and independence in their professional judgment.

Management - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

Profession - Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

Colleagues - Software engineers shall be fair to and supportive of their colleagues.

Self - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Problems With Professional Codes

- Can be treated like a professional algorithm
- Assumption that ethical problems can be solved *simply* by referring to codes
- Lack of understanding of the general moral principles underlying codes
- Can lead to a lack of ethical debate
- Misunderstanding that codes are mere end products of ethical deliberation
- The fallibility of codes and need for improvement
- Limitation in scope and potential for conflict with other codes

Comparison of Different Codes

- BCS is essentially *deontological*
  - Obligation to prevent harm
- ACM has a *utilitarian* emphasis
  - Enhancing end-user welfare
- Neither offers a complete guide to all situations (and there are many other guides we can follow)
- Can you follow both?
  Which code would the professional who is a member of both bodies follow when they come into conflict?

Other Issues

- What else is required to make the code of conduct useful?

Summary

- The need for professionalism in IT
- The professional bodies
- Codes of Conduct and Practice